

Ricardo Andrés Calvo Méndez

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SUMMARY

Ph.D. student in **formal verification, program analysis, software security** and **performance optimization**, with **3+ years of industry experience** in QA and full-stack development. Seeking **Summer 2027** opportunities.

EDUCATION

Ph.D. in Electrical and Computer Engineering 2025–2030 (expected)
Purdue University, West Lafayette, IN
B.Sc. in Systems and Computer Engineering 2018–2023
Universidad Nacional de Colombia, Bogotá, Colombia

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Research Assistant Aug 2025–Present
Purdue University, West Lafayette, IN

- Co-develop AutoSOUP, an LLM-based system that automates memory-safety verification of C code (CCS 2026).
- Designed its **debugging agent**, which diagnoses failed proofs and proposes fixes; set up cluster evaluations.
- It exposed **66.7% of 60 known CVEs** (vs. 28.3% for a coding agent) and found **20 new vulnerabilities**.

Test Engineer & QA Analyst Sep 2024–Jul 2025
Kravata SAS, Bogotá, Colombia

- As the **first and only QA engineer**, defined the company's testing strategy and standards from scratch.
- Automated unit, end-to-end and CI/CD tests for the **4 most critical services** with Cucumber and Selenium.

Full-Stack Software Developer Aug 2023–Aug 2024
Junior Software Developer Aug 2021–Jan 2023
Fit Ideas SAS, Bogotá, Colombia

- Led **4 developers** on a **white-label** fuel sales app: defined architecture, reviewed code, supported Scrum.
- Built web and mobile apps (Spring Boot, React, Django, Flutter) and a configurable public-data web scraper.

Undergraduate Visiting Scholar (UREP-C) Feb 2023–Jul 2023
Purdue University, West Lafayette, IN

- With mentorship, designed and implemented EmNetTest's fuzzing strategy and its evaluation (ASE 2023).
- Helped triage results: reproduced **12 known vulnerabilities** and found **7 new defects (6 CVEs)**.

TEACHING

Graduate Teaching Assistant, ECE 30861: Software Engineering Fall 2026
Purdue University, West Lafayette, IN

SELECTED PUBLICATIONS

- H. Peng*, **R. Calvo***, et al. *How Do Coding Agents Optimize Software and Report Performance Validation? A Large-Scale Empirical Study of Open-Source Pull Requests* [arXiv](https://arxiv.org/abs/2601.00000). Preprint 2026. *Equal contribution.
- P. Amusuo, **R. Calvo**, et al. *AutoSOUP: Safety-Oriented Unit Proof Generation for Component-level Memory-Safety Verification* [arXiv](https://arxiv.org/abs/2601.00000). ACM CCS 2026. Presenting author.
- P. Amusuo, **R. Calvo**, Z. Xu, A. Machiry, and J.C. Davis. *Systematically Detecting Packet Validation Vulnerabilities in Embedded Network Stacks* [DOI](https://doi.org/10.1145/3588888). ASE 2023.

SKILLS

Research & Security: Formal verification, model checking (CBMC), program analysis, fuzzing, performance optimization
Programming: C, C++, Python, Java, JavaScript, TypeScript, Dart, SQL
AI for SE: LLM agents, agentic pipelines, empirical evaluation
Engineering: Spring Boot, React, Django, Flask, Flutter, PostgreSQL, Selenium, Cucumber, CI/CD, Docker, Linux, Git